MILITARY SPECIFICATION

VIEWER/PRINTER, MICROFICHE (24X AND 48X)

This amendment forms a part of Military Specification MIL-V-80241A, dated 3 April 1974, and is approved for use by all Departments and Agencies of the Department of Defense.

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3.3.1.4, delete and substitute:

"3.3.1.4 Size. Dimensions shall be such as to permit passage through a rectangular doorway 20 inches wide."

3.3.1.5, delete "120 pounds" and substitute "180 pounds."

3.3.1.6, delete and substitute:

"3.3.1.6 Hood. Rear projection V/P shall be equipped with a hood which extends toward the operator a minimum of 6 inches from the top of the screen or, when an incline screen is used, 6 inches in front of the protruding point of the screen; and be tapered on the sides to approximately 1 inch at the bottom of the screen. The interior shall be light absorbing black."

3.3.2, line 3, delete "14"and substitute "11".

*3.3.3.2, line 1, delete "500" and substitute "250".

3.3.6, line 1, delete "(24X and 48X)" and substitute "used".

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3.3.6.2, line 2, delete "and its focal length" and add a "period" after lens

3.3.6.3, first sentence, delete "48X ±4 percent." and substitute "from 36X to 48X ±4 percent."

3.3.6.6, line 2, delete "24X and 48X magnification" and substitute "the two magnifications".

3.3.7, delete paragraph in its entirety.

3.3.8.1, delete and substitute:

"3.3.8.1 Carrier design. Fiche carrier mechanisms that require opening and closing for insertion or removal of fiche shall be self opening and remain open when pulled toward the operator from any horizontal position of the carrier and self closing when pushed in away from the operator. This operation shall open the front of the platens to a minimum of 1 inch with a maximum of 2 pounds pull, and close the platens with a maximum of 2 pounds push. The carrier mechanism may be front or side loading, and it shall be designed for convenient operation. The carrier shall accomodate a $105.00^{+0.00}_{-0.75}$ x $148.00^{+0.0}_{-1.0}$ mm fiche (see figure 1), and provide for viewing of the data area plus a 0.80 mm border. When platens are used, a means shall be provided for loading and unloading fiche from the carrier without scratching or otherwise damaging the fiche, and arranged so all surfaces of the platen may be cleaned. If transparent platens are used, they shall be scratch resistant."

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3.3.9.2, delete and substitute:

"3.3.9.2 Print paper processing unit. The processing unit required for developing the exposed print paper shall be designed to prevent chemicals/toner, if required, from spilling, contaminating each other, or in any way becoming free from the processing unit and contaminating, soiling, or damaging any part of the V/P to a maximum of 5 degrees tilt from the vertical in any direction when in operation. The complete image of the print produced shall be a blowback of not less than 71 percent of the 48:1 microimage or 92 percent of the 24:1 microimage."

3.3.9.4, delete and substitute:

"3.3.9.4 Print paper feed. The equipment shall be motorized to move paper automatically from the paper storage area through processing and deliver a finished cut print within the reach of the operator's position. V/P's shall be equipped with an automated paper feed that will deliver printed sheets either full size or, at the operator's option, at approximately one-half size without reducing the paper width, and without changing or reloading the paper supply."

3.3.9.5, add to end of paragraph: "Paper cutter shall not rust or peel under normal use."

3.3.10, delete and substitute:

"3.3.10 Printing material storage. The printing processing compartment shall be designed to enclose paper and chemicals, if required, to prevent deterioration of such material within the V/P. Paper stored in the V/P shall be capable of producing a suitable print as defined in 3.3.12.1 for a period of 6 months when the V/P is utilized in an environment with maximum temperature of 75 degrees fahrenheit and relative humidity not more than 50 percent."

3.3.11.2, line 2, delete "the lower one-fourth of the V/P on".

3.3.12 Delete second sentence and substitute the following:

"Cost per printout is an evaluation factor and shall be included in award evaluation (see 6.2)."

3.3.13, line 4, delete "viewers" and substitute "V/P's".

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3.4.1, delete "60" and substitute "90"

3.4.2, delete "+115°F" and substitute "+105°F"

3.5, line 2, change "printed" to "printer".

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3.12.2, delete paragraph in its entirety.

3.14, delete and substitute:

"3.14 Operating and maintenance manuals. Operating and maintenance manuals (separate or combined) shall be furnished with each V/P (see 6.8)."

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Add new paragraphs:

"3.16 Durability.

3.16.1 Durability test for carrier. When tested as specified in 4.17.1, the carrier shall remain mechanically operative and shall meet the requirements of 3.3.8.1.

3.16.2 Durability test for viewing and printing. When tested as specified in 4.17.2, each control shall function properly and meet the requirements of 3.5."

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Add to TABLE I below "Vibration":

"Durability - - - - - - - - - - - 3.16 4.17 "

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Add to TABLE III below "V/P operating test":

"Durability - - - - - - - - - - - 3.16 4.17 0 0 1 per 500 units .".

*4.8, line 2, delete "+115°F" and substitute "+105°F".

4.10, delete and substitute:

"4.10 Resolution of fine detail (See 3.5). The V/P when properly focused, with 24X lens or 36X to 48X lens, on a film target per figure 2, reduced 24X or 36X to 48X respectively, shall resolve 3.2 lines/mm or higher on the viewing surface in all four corners and the center without refocusing. (Compromise focusing may be used.) The reduced film target shall be certified to have a minimum resolving power of 108 lines/mm and 216 lines/mm respectively and a bar to space width tolerance of ±10 percent when viewed with a 100X or 200X microscope, respectively."

4.12, line 2, delete "class II B" and substitute "class 1 C".

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4.15.1, fourth sentence, delete "fully enlarged to the 24X and then the 48X magnification" and substitute "100 ±8 percent for 24X and a minimum of 71 percent for 48X fiche, as".

4.16, delete and substitute:

"4.16 Vibration (see 3.11). The V/P shall be tested in accordance with MIL-STD-167, Type I, except chemicals and lamp may be removed and the carrier may be secured during test."

Add new paragraphs:

"4.17 Durability.

4.17.1 Durability test for carrier. The carrier shall be installed in the V/P and fully cycled for 8,000 times from right to left, 8,000 times from front to rear, and 8,000 times from load to view to unload positions. The front to rear and load to view to unload tests may be cycled in one operation. Any failure shall stop the test. The carrier may be reworked and life test restarted at the beginning.

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4.7.2 Durability test for viewing and printing. The V/P shall be cycled for 5,000 operations. The exposure control shall be set in the mid range (when external exposure control is used). Any failure (except for lamps or fuses) shall stop the test. The V/P may be reworked and life test restarted at the beginning."

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6.2, item (c), delete and substitute:

"(c) Cost per printout will be based on the cost per full size print (paper and chemicals), computed using the current price of the paper identified for use by the bidder/offerer for his equipment as reflected on the current published price list for 1 roll or box, if cut paper is used, multiplied by 22,000. This composite cost will be added to the unit price of the equipment for evaluation purposes only (see 3.3.12)."

*6.2, add "e"

"(e) Installation and maintenance service requirements; when required."

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6.5, delete and substitute:

"6.5 Operating and maintenance manual. Two copies of the manufacturer's service manual for maintenance and repair purposes should be provided with each V/P. This manual shall include a complete list of repair parts and a list of recommended running spares to support the initial year of operation (see 3.14)."

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